W-F-391c
November 27, 1963

SUPERSEDING
Int. Fed. Spec. W-F-00391b (GSA-FSS)
September 4, 1962 and
Fed. Spec. W-F-391a
October 26, 1945

FEDERAL SPECIFICATION

FIRE ALARM SYSTEM; LOCAL, ELECTRIC, SUPERVISED, SUCCESSION CODED, POSITIVE NONINTERFERING TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers electric, supervised fire alarm systems of the succession coded, positive noninterfering type for the purpose of evacuating public buildings and alerting cognizant fire-fighting personnel through the sounding of a coded signal.

1.2 **Classification.**

1.2.1 *Systems.*

1.2.1.1 *Type, classes, styles, and kinds.* Fire alarm systems furnished under this specification shall be of one type (succession coded) and the following classes, styles, and kinds, as specified (see 6.1).

Classes:

Class 1—General alarm.
Class 2—Presignal alarm.

Styles:

Style A—Manual.
Style B—Combination automatic-manual.

Kinds:

Kind A—Nonrecording.
Kind B—Recording with time stamp.

1.2.2 *Stations.*

1.2.2.1 *Type, classes, and styles.* Fir alarm stations furnished as a part of the fir alarm system shall be of one type (positiv noninterfering and succession coded) an the following classes and styles, as specifie (see 6.1).

Classes:

Class 3—Break-glass station.
Class 4—Nonbreak-glass station.

Styles:

Style C—Interior use.
Style D—Exterior use.
Style E—Special purpose.

1.2.3 *Automatic fire detectors (style 1 systems only).*

1.2.3.1 *Types.* Automatic fire detector furnished as a part of style B fire alarn systems shall be of the following types, a specified (see 6.1).

Type I—Fixed temperature, spot type nonrestorable.
Type II—Fixed temperature, spot type restorable.
Type III—Fixed temperature, line type nonrestorable.

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Type IV—Rate-of-rise, line type, restorable.
Type V—Rate-of-rise, spot type, restorable.
Type VI—Rate-of-rise, spot type, restorable (thermo-electric principle).
Type VII—Rate-of-rise, line type, restorable; fixed temperature, spot type, nonrestorable.
Type VIII—Rate-of-rise, spot type, restorable; fixed temperature, spot type, nonrestorable.
Type IX—Rate-of-rise, spot type, restorable; fixed temperature, spot type, restorable.

Note: For definition of the various technical terms used throughout this specification, reference should be made to NEMA Standard SB1-1962.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

J-C-129—Cable and Wire, Thermoplastic-Insulated, General Purpose (0 to 600-Volt Service).
J-C-145—Cable and Wire; Weather-Resistant.
L-P-507 — Plastic Sheet, Extruded Acrylic.
W-B-134 — Battery, Storage (Lead-Acid, Industrial, Floating Service).
GG-T-406—Time Stamps (Correspondence).
WW-C-563—Conduit, Metal, Rigid; & Bend & Elbow, Electrical Conduit: Thin-Wall Type (EMT).
WW-C-571—Conduit, Metal, Rigid; and Coupling, Elbow, and Nipple, Electrical Conduit: Enameled.
WW-C-581—Conduit, Metal, Rigid; and Coupling, Elbow, and Nipple, Electrical Conduit: Zinc-Coated.

Federal Standards:

Fed. Std. 102 — Preservation, Packaging, and Packing Levels.
Fed. Std. 123 — Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standard:

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Underwriters' Laboratories, Inc., Standards:

UL 464—Audible Signal Appliances.
UL 521—Fire-Detection Thermostats.
UL 864—Control Units for Fire-Protective Signalling Systems.

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(Copies may be obtained from the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, N. Y. (11749); 207 East Ohio Street, Chicago 11, Illinois; or 1655 Scott Blvd., Santa Clara, California.)

American Standards Association, Inc., Publication:

C39.1 — Electrical Indicating Instruments, (Panel, Switchboard, & Portable Instruments), Requirements for.

(Copies may be obtained from the American Standards Association, Inc., 10 East 40th Street, New York 16, New York.)

National Electrical Manufacturers' Association Publication:

SB1-1962—Signalling Apparatus, Local Fire Alarm Systems.

(Copies may be obtained from the National Electrical Manufacturers Association, 155 East 44th Street, New York 17, N. Y.)

National Board of Fire Underwriters Publication:

Std. No. 72—Protective Signalling Systems; Proprietary, Auxiliary, and Local; Installation, Maintenance, and Use of.

(Copies may be obtained from the National Board of Fire Underwriters', 85 John Street, New York 38, New York; 222 West Adams Street, Chicago 6, Illinois, or 465 California Street, San Francisco 4, California.)

3. REQUIREMENTS

3.1 The design, material, workmanship, finish, and operation of the system and all its parts shall be adequate for the purpose intended, of the manufacturer's current production, and in full accordance with the applicable provisions of NBFU Standard No. 72.

3.2 Fire and casualty hazards.

3.2.1 Each bidder shall submit to the contracting agency proof that the fire alarm system components he proposes to supply under this specification, conform to the requirements of UL-464, UL-521, and UL-864, as applicable. The label, or listing with reexamination, of the Underwriters' Laboratories, Inc., may be accepted as evidence that the components conform to this requirement.

3.2.2 In lieu of the Underwriters' Laboratories, Inc., label or listing with reexamination, the bidder may submit independent proof satisfactory to the contracting agency that his fire alarm system components conform to the requirements, including methods of tests, of the Underwriters' Laboratories, Inc.

3.2.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.3 Systems.

3.3.1 *General.* All systems shall be of the double supervised, succession coded type equipped with electric, single-stroke or vibrating bells, as specified (see 6.1). Other signalling devices, such as horns, annunciators, (including the graphic type), etc., shall be furnished when specified in the invitation for bids. The alarm initiating and indicating devices, control units, etc., shall constitute a unified system for the satisfactory operation of which one manufacturer shall be responsible. The name or trademark of the manufacturer shall appear on all major components. When specified in the invitation for bids, other protective signalling systems shall be combined with and made to operate in conjunction with the fire alarm systems described herein.

3.3.1.1 The alarm initiating device mechanism and circuitry shall be so arranged that should two or more devices be operated at or near the same time, the device first securing the line shall continue to transmit its

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code without interference from any other device. The remaining actuated devices shall similarly transmit their codes without interference as the line becomes available.

3.3.2 *Power.* Except as otherwise provided herein, power supply sources shall conform to the applicable requirements of NBFU Standard No. 72, Article 220.

3.3.2.1 *Operating power.* The systems shall be operated from either a 208/120- or 120/240-volt, 60-cycle, unfused grounded neutral, alternating-current service directly; or through a rectifier storage battery combination, as specified in the invitation for bids (see 6.1). The trickle-charged, floating storage battery shall operate the system in the event of an alternating-current supply failure.

3.3.2.2 *Supervisory power.*

3.3.2.2.1 *A-C operation.* The supervisory power for the operation of the audible and visual trouble-indicating devices shall be obtained from the grounded neutral and the phase wire not used for the operating power. The supervisory arrangement shall be such that an audible warning shall be given upon failure of either the operating or the supervisory power. When heavy duty signalling devices operated from an auxiliary source are specified, the power source thereof shall be supervised.

3.3.2.2.2. *D-C operation.* The supervisory power for the operation of the audible and visual trouble-indicating devices shall be obtained from the stand-by storage battery. The supervisory arrangement shall be such that an audible warning shall be given upon failure of either the normal operating rectified power or the supervisory power. When heavy duty signalling devices operated from an auxiliary source are specified, the power source thereof shall be supervised.

3.3.3 *Class 1 systems.*

3.3.3.1 *Operation.* The operation of any alarm initiating device shall cause its code to sound on all alarm bells and other sounding devices in the system.

3.3.3.1.1 When so specified (see 6.1), the system shall be provided with a feature to sound a continuous alarm after the selective coded alarm signal has been completed to provide an evacuation signal of longer duration.

3.3.4 *Class 2 systems.*

3.3.4.1 *Operation.* The operation of any alarm initiating device shall cause its code to sound only on the presignal bells. Provision shall be made in each station for sounding a general alarm by inserting a key or plug with a special shaped shank, particularly identified for the purpose, into a keyhole or receptacle before operating the station. The key or plug, while inserted, shall be plainly visible or shall prevent the door, if so equipped from closing. When other jacks (telephone, etc.) are provided in the station, each jack shall be properly identified to avoid confusion in operation.

3.3.4.1.1 When so specified (see 6.1), the system shall be provided with a feature to sound a continuous alarm after the selective coded alarm signal has been completed to provide an evacuation signal of longer duration.

3.3.4.2 *General alarm switch.* When specified in the invitation for bids, a switch shall be provided at the control unit whereby any alarm transmitted can be made to sound all alarm bells and other signalling devices in the system. A red pilot lamp or other signal at the control unit shall indicate when the switch is in the general alarm position. When specified in the invitation for bids, additional signals indicating the position of the switch shall be furnished.

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3.3.5 *Style A systems.* Style A systems shall consist entirely of manually operated stations and the necessary circuitry and hardware required for an efficient and integrated system as hereinafter provided.

3.3.6 *Style B systems.* Style B systems shall consist of both manually operated stations and automatic fire detectors and transmitters for the protection of both habitable and storage areas. Automatic detector transmitters shall provide a coded signal indicating the area in which the actuated detector is located.

3.4 **Stations.**

3.4.1 *General.* Except as otherwise provided herein, stations shall conform to the applicable requirements of NEMA Standard SB1-1962, Part 4, Section 1.

3.4.1.1 All stations shall be of the pull-lever type. The act of pulling the lever shall either wind an operating spring in the station mechanism which, when the lever is released, shall drive the mechanism, or energize an electric motor-driven device to sound at least four rounds of the code. The design shall be such that after the lever has been pulled and released, it shall return to its initial position and either trip free from the mechanism or be locked in its initial position so that it shall be impossible to interfere with or confuse the signals by further manipulation of the lever. The arrangements shall be such that if the lever is not pulled all the way down, the station shall transmit at least a proportionate number of rounds, but in no case shall it transmit only a portion or a round.

3.4.2 *Test facilities.* Means shall be provided for testing the station mechanism without transmitting any fire alarm signal. Provisions shall be made so that the system cannot be left inoperable with the test key removed. When test keyhole is normally open to view, it shall not be possible to make the station inoperable by inserting and leaving a small object (not the test key) in the test keyhole. The construction of the test key feature shall be such that finger springs or other delicate parts cannot be put out of adjustment by undue forcing of the key. Means shall be provided for transmitting individual taps to all alarm bells without transmitting any fire alarm signal.

3.4.3 *Construction.*

3.4.3.1 *Class 3 stations.* Class 3 stations shall be equipped with a glass plate or rod with breaking means secured to the station and so constructed that either pulling the pull lever will break the glass and actuate the station, or that breaking the glass will provide access to the pull lever. Provision shall be made for authorized personnel to circumvent the glass breaking feature to facilitate test and inspection procedures.

3.4.3.2 *Style C stations.* Stations designed for interior use shall have a cast metal or galvanized, bonderized, or equivalent sheet-steel back box, shall be arranged for threaded metal conduit connections, or knockouts, at the top and bottom, and shall have an inner case of substantial protective metal, plastic, or glass enclosing the mechanism, when applicable. Sheet-steel back boxes shall be of metal having a nominal thickness of No. 16 gauge. All openings into the inner enclosure shall be substantially dusttight. The mechanism shall be housed in a manner to permit easy access to the movement, by means of a special tool, so that it can be inspected without disconnecting the station from its circuit or putting tension on the station leads. A terminal block of phenolresin composition, porcelain, or equivalent, insulating material shall be provided to readily identify and facilitate connection of the station leads to the line. The outer case of the station shall extend not more than 2-1/2 inches beyond the finished wall unless surface mounted stations are specified.

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3.4.3.2.1 *Number plates.* A number plate, with raised or depressed figures corresponding to the code number of the station, shall be secured to the outer door, if so constructed, or where it may be seen if no door is used. All number plates shall be interchangeable.

3.4.3.2.2 *Finish.* Unless otherwise specified in the invitation for bids, all exposed metal parts of enclosing cases (except number plates and small trims) shall be given a priming coat and one or more finishing coats of red lacquer, English vermillion, or red enamel to produce a smooth, hard, durable finish. Suitable directions for operating the station shall be provided on the cover in raised or depressed white enameled or aluminum letters. All surface back boxes shall be given a priming coat of iron oxide and one coat of red lacquer or an equivalent protective coating.

3.4.3.3 *Style D stations.* Stations designed for exterior use shall comply with the applicable requirements for style C stations and the entire mechanism shall be housed in a cast metal, weatherproof case with hinged door and catch, and a suitable door pull. A suitable approved pedestal shall be provided when specified.

3.4.3.4 *Style E stations.* Special purpose stations shall be of cast bronze or other material, as specified, in the invitation for bids. Such stations shall otherwise meet all requirements of this specification and the special requirements of the invitation for bids concerning appearance, finish, etc. Unless otherwise specified, exposed parts shall be of a finish and color to match other metal work in the vicinity.

3.4.3.4.1 All parts shall be of high quality as to material and workmanship. The station submitted by the manufacturer shall be specifically approved prior to shipment. The stations shall project not more than 2-1/2 inches beyond the finished wall. When so specified, station enclosures shall be furnished which do not project more than 5/8 inch from the finished wall.

3.4.3.4.2 The pull lever shall be protected by an outer door and shall be of substantial construction, rust-proof and polished. The lever shall be nickel-plated or finished to harmonize with the surrounding parts of the lever well. Raised lettering of uniform height giving instructions for operation shall be cast upon the door and lever well, such letters to be 1/4 inch or larger, clear cut, flat-faced, polished and lacquered. No other lettering except the code number plate, shall appear upon the face of the station.

3.5 Automatic fire detectors.

3.5.1 *General.* Unless otherwise specified, automatic fire detectors shall conform to the applicable requirements of UL 521 and NEMA Standard SB1.1962, Part 4, Section 2.

3.6 Transmitters.

3.6.1 *General.* Except as otherwise provided herein, transmitters shall conform to the applicable requirements of NEMA Standard SB1-1962, Part 4, Section 3.

3.6.2 *Operation.* Transmitters shall be of the electrically-tripped type, either motor, spring, or impulse driven, and shall transmit a coded alarm signal, corresponding to the location of the detectors connected thereto, of at least four rounds. Once actuated for alarm, the transmitter shall complete its signal without interruption due to either the automatic or manual restoration of the actuated detector during signal transmission. When motor driven transmitters are supplied, the motor shall be connected to a supervised circuit in the control unit. When spring driven transmitters are supplied, they shall be manually wound and shall be provided with a manually operable feature in addition to the primary automatic function. Such units shall require rewinding

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after each complete operation and shall sound an audible trouble signal when in the run-down condition.

3.7 **Alarm indicating devices.**

3.7.1 *General.* Except as otherwise provided herein, alarm indicating devices shall conform to the applicable requirements of UL 464 and NEMA Standard SB1-1962, Part 5.

3.7.2 *Alarm bells.*

3.7.2.1 *General.* Alarm bells shall be of the electric, single-stroke or vibrating, no-contact, under-dome type, as specified (3.3.1). The operating mechanism shall be rustproof, suitably protected from dust and insects, and housed behind the gong shell.

3.7.2.2 *Finish.* Unless otherwise specified in the invitation for bids, exposed metal parts, other than the gong shell and nut, shall be given a suitable priming coat, and one or more finishing coats of the manufacturer's standard color to produce a smooth, hard, durable finish.

3.7.2.3 *Gong shell.* The gong shell shall be made of high-grade hot-pressed alloy steel or of bell metal. The outside of the gong shell and the securing nut shall be smooth and shall be given an approved rustproof finish. Unless otherwise specified in the invitation for bids, general alarm bells shall be 10 inches in diameter and presignal bells shall be 6 inches in diameter.

3.7.2.4 *Recessed mounting.* When so specified in the invitation for bids, alarm bells shall be furnished mounted to a 4-inch or 4-11/16-inch square outlet box firmly fastened to the inside rear wall of a flush wall shell fabricated from sheet steel or aluminum. Sheet steel boxes shall have a nominal thickness of No. 16 gauge. Sheet aluminum boxes shall have a nominal thickness of No. 12 gauge. Wall shells shall be provided with two combination 1/2- and 3/4-inch knockouts at both top and bottom. Flush grille covers shall be fabricated from aluminum or bronze, or as specified in the invitation for bids, to harmonize with other materials and devices in the area, and shall have a nominal thickness of No. 10 gauge. The grille covers shall mount to the shells with four mounting screws of the same material as the grilles.

3.7.2.5 *Outdoor service.* Outdoor alarm bells shall, in addition to conforming to the above requirements, be of an approved weatherproof type, suitably mounted, and protected by a nonferrous metal housing with a grille or louvers.

3.7.3 *Audible trouble signals.* Unless otherwise specified in the invitation for bids, audible trouble signals shall be of the vibrating type, capable of producing a tone distinctive from the tone of other audible signals.

3.7.4 *Other audible signalling devices.* When other audible signalling devices are specified to be furnished in addition to or in lieu of the above devices, they shall be capable of operating together satisfactorily. Where horns are used in place of bells, they shall be designed for the same type of supervisory and operating circuitry as the bells, except for d.c.-operated horns (see 3.12), where operation and supervision shall comply with NBFU Standard No. 72, paragraph 2424a.

3.8 **Control units.**

3.8.1 *General.* Except as otherwise provided herein, control units shall conform to the applicable requirements of UL 864 and NEMA Standard SB1-1962, Part 3.

3.8.2 *Wiring.* All wiring shall be on the rear of control unit panels, securely fastened in place, and arranged in neatly served cables. External wiring shall terminate on terminal strips mounted within the cabinet. Wire shall conform to the requirements for type THW of J-C-129.

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3.8.3 *Control board.* The control board shall be of the unit panel or modular type construction and shall have mounted thereon all the necessary switches, meters, relays, pilot lamps, and resistors. It shall be installed in a surface or flush, wall-mounted, steel enclosure; free-standing, floor-mounted console; or desk assembly as specified in the invitation for bids. All operations required for the various tests or normal functioning or care of the system shall be performed on the front of the board.

3.8.3.1 *Panels.* Panels shall be of high electrical quality asbestos composition, or approved equivalent, shall have a high-grade finish, and shall test free from electrical and mechanical defects. Approved modular metal construction is acceptable.

3.8.3.2 *Resistors.* All rheostat, balancing, or other resistors mounted on the panels shall be enclosed in such a manner as to prevent accidental contact by maintenance personnel, unless the location is such that line terminals and testing points are sufficiently removed from the resistors that accidental contact is unlikely in the course of maintenance work. Resistors shall be readily accessible when necessary for test or replacement. In addition, resistors shall be located in such a manner as to preclude undue heating of the immediate area. Lamps shall not be used as resistors.

3.8.3.3 *Relays.* All relays shall have self-cleaning contacts of ample weight silver or alloy of equivalent performance. Supervisory relays shall be suitably protected against dust.

3.8.3.4 *Instruments and soldered joints.* Instruments shall conform to ASA Standard C39.1 and shall be of high grade both as to materials and workmanship. All soldered connections shall be mechanically secured before soldering.

3.8.4 *Enclosures.*

3.8.4.1 *General.* Enclosures shall be provided with ample gutter space to allow for proper clearance between the enclosure and live parts of the panel equipment. Enclosures shall be substantially dusttight and shall have one or more doors containing glass inserts, where necessary, to permit visual observation of instruments, pilot lamps, etc. Enclosures with doors over 48 inches high shall be provided with a 3-point catch and lock; all other doors shall contain a cabinet type cylinder lock.

3.8.4.2 *Rear door cabinets.* When so specified, the cabinet shall be installed in the partition wall where access to the back of the cabinet may be gained from the reverse side of the partition. In such locations the back of the cabinet shall be suitably trimmed and shall have steel doors to give full access to all wiring on the back of the panel.

3.9 **Circuits.** The number of circuits and loops, and the number of stations, detectors, transmitters, and alarm bells shall be as specified in the invitation for bids. Alarm initiating devices shall be connected in one circuit and alarm indicating devices in another. Circuits shall consist of one or more loops, each of which shall originate and terminate at the control unit. There shall be the same number of general alarm loops as there are alarm initiating loops. In class 2 systems, the general alarm control shall have the same number of general alarm control loops as there are presignal alarm initiating loops. Recorders, when used, shall be connected on a separate circuit and shall record all alarm signals sent from any initiating device. The system shall be capable of operating satisfactorily on from 85 percent to 110 percent of normal voltage.

3.9.1 *Testing.* Facilities shall be provided whereby any alarm initiating or alarm indicating loop (including the "general alarm control loop" in class 2 systems) may be taken out of the system for test or repair. The act of removing any loop shall automati-

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cally leave the remainder of the circuit in normal operating condition and light a pilot lamp, which shall remain lighted until the loop is restored to normal.

3.9.2 *Supervision.* The system shall be so arranged that a single open or a single ground (except a ground that does not interfere with the transmission of an alarm) or other abnormal circuit condition in any part of the system, including the rectifier and battery on d-c systems, shall cause the trouble bell to ring continuously. A switch shall be provided which, when operated, shall silence the bell and light a pilot lamp until the trouble is cleared and the system restored to normal, at which time the trouble bell shall sound until the switch is restored to its normal position. Restoration of the switch shall silence the bell and extinguish the pilot lamp.

3.9.2.1 *Alarm initiating circuit.* The supervision of the alarm initiating circuit shall include the coding contacts of these devices, except contacts connected in multiple across electrically supervised circuits.

3.9.2.1.1 *General alarm control circuit (class 2 systems).* The supervision of the general alarm control circuit shall include all general alarm contacts in series.

3.9.2.2 *Alarm indicating circuit.* The supervision of the alarm indicating circuit shall include the circuit through the alarm bell windings.

3.9.3 *Operation on open or grounded circuit.* When specified in the invitation for bids, the station circuits shall be arranged so that an open circuit or a ground, or an open circuit and a ground occurring at the same point, shall not prevent the transmission of a clear, intelligible signal before the trouble is cleared. This feature shall be accomplished entirely by automatic or entirely by manual operation, as specified (see 6.1).

3.10 **Recorders.** The recorder shall automatically make a permanent record of all alarms sent in by inked imprint or triangular or circular punchings, corresponding to the code, on a paper tape. The recorder shall function properly at any rate up to three blows a second. The recorder shall consist of a spring-driven or a motor-driven mechanism which, on receipt of signals, shall drive the tape forward at a uniform rate and automatically stop the tape at a predetermined time after the last complete round of signals has been received. The punch die, if so equipped, shall be of hardened tool steel. Tape reels shall be so arranged that new rolls of tape can be placed in position without the use of tools. The recorder circuit need not be supervised.

3.10.1 *Enclosure.* The driving mechanism shall be enclosed in a dusttight plate glass, plastic, or finished metal case, which shall be easily removable for access to the working parts.

3.10.2 *Take-up reel.* The take-up reel shall be of the quality and finish as specified for the recorder. It shall maintain a uniform tension and keep the tape taut regardless of the size of the roll. It shall be capable of being wound while the signals are being received without interfering with its normal operation. If spring-driven, an automatic brake shall be provided to instantly stop the mechanism in the event of breakage of the tape. It shall permit the tape to be withdrawn, if desired, to inspect the preceding recorded signals. Working parts shall be protected from dust. The paper roll shall be easily removable without the use of tools.

3.10.3 *Time stamp.* The time stamp shall be electrically operated and either of the synchronous motor or the impulse type conforming to GG-T-406, type I. The stamping element shall be controlled from the recorder.

3.10.4 *Mounting.* The recorder, take-up reel, and time stamp shall be either one en-

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closed unit or the separate units shall be housed under a plate glass or plastic cover with a suitable lock.

3.11 Graphic annunciators. When so specified (see 6.1), the system shall be provided with a graphic annunciator consisting of a reverse-engraved plastic sheet, conforming to the requirements of L-P-507, of 3/16 inch minimum thickness, and color-filled to represent building outlines, floor plans, area configuration, etc., lettering, symbols, and other details pertinent to the system. The plastic panel shall be attached by suitable means to the control board. Each alarm initiating device in the system shall be represented by a pilot light on the annunciator and shall be identified as such. Actuation of an alarm initiating device shall cause its associated pilot light to light and indicate the location of the actuation. The pilot lights shall be enclosed in surface or flush-mounted plastic shields (domes) and shall be attached to the annunciator by means of threaded inserts through the plastic sheet. All electrical connections to the pilot lights shall be behind the steel panel on which the annunciator is mounted. The electrical design of the annunciator shall be such that resetting a specific alarm or trouble signal will not reset any other alarm or supervisory indication still in effect.

3.12 Storage battery and rack (when used). The storage battery shall conform to the requirements of W-B-134, class 1, shall consist of the necessary number of cells for normal requirements, plus four additional cells and shall have sufficient capacity to operate the system for 60 hours in the event of failure of the rectified supply. Operation of the system shall be possible on 80 percent of the normal battery output. The cells shall be furnished with all necessary intercell connectors, terminal lugs, bolt connectors, and hydrometer syringe. The rack shall be of metal step-type design, of ample size and strength to accommodate the number of cells furnished, and shall be suitably coated to resist corrosion.

3.13 Rectifier and charging panel (when used).

3.13.1 *Rectifier.* The rectifier shall be complete with transformer and regulating reactor for supplying current to the system and shall provide a floating charge sufficient to maintain the battery in a fully charged condition. The primary winding of the transformer shall be arranged with taps for connection to the alternating-current supply of 105 to 125 volts. The secondary of the transformer shall be provided with the necessary taps so that after rectification, the required voltage shall be available. A complete spare rectifier shall be furnished for each installation.

3.13.2 *Charging panel.* The battery charging panel shall contain all necessary apparatus for proper trickle and booster charge of the complete battery unit. The equipment shall include a main line switch, fuses, an ammeter and a voltmeter to indicate battery charging rate, and a current regulating switch. The voltmeter shall have a red arrow on the dial to indicate the normal trickle charge voltage of the battery. The charging panel shall be enclosed in a steel cabinet having a hinged door with glass or plastic inserts to provide a clear view of meters and other essential equipment when the door is closed.

3.14 Performance.

3.14.1 *Alarm initiating devices.*

3.14.1.1 *Stations and transmitters.* Unless otherwise specified (see 6.1), the following requirements shall conform to NEMA Standard SB1-1962, Part 4, Section 1:

(a) Normal operation
(b) Contact overload
(c) Endurance
(d) Vibration
(e) Jarring
(f) Dielectric strength
(g) Insulation resistance

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(h) Spray (style D stations only)
(i) Humidity (style D stations only)

3.14.1.2 *Detectors.* Unless otherwise specified (see 6.1), the following requirements shall conform to UL 521:

(a) Calibration
(b) Sensitivity
(c) Premature operation
(d) High temperature exposure
(e) Corrosion
(f) Paint loading
(g) Contact overload
(h) Endurance
(i) Vibration
(j) Rough usage
(k) Dielectric strength

3.14.2 *Alarm indicating devices.* Unless otherwise specified, the following requirements shall conform to UL 464.

(a) Current rating
(b) Endurance
(c) Temperature rise
(d) Dielectric strength
(e) Spray (outdoor service devices only)
(f) Humidity (outdoor service devices only)

3.14.3 *Control units.* Unless otherwise specified, the following requirements shall conform to UL 864:

(a) Normal operation
(b) Electrical supervision
(c) Input and output current
(d) Jarring
(e) Temperature rise
(f) Contact overload
(g) Endurance
(h) Dielectric strength
(i) Abnormal operation
(j) Burnout

3.15 **Installation.** The fire alarm system, including conduit, tubing, and wiring, shall be completely installed by the contractor and all wires shall be properly tagged. The installation shall comply with the applicable requirements of NBFU Standard No. 72. All interior wiring, including conductors at ground potential, shall be type THW insulated copper wire not smaller than AWG 14. All exterior aerial wiring shall be of the weather-resistant type conforming to J-C-145, not less than AWG 10. Conduit shall conform to WW-C-571, WW-C-581, WW-C-563, or as applicable. Conduit shall not be used for ground return.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Where, in the course of manufacture, the quality controls exercised by the manufacturer or his suppliers in the production of materials or components used in the system, completely obviate any need for any particular inspection or test procedure specified herein, the pertinent quality control records may be furnished in lieu of the performance of the inspection or the test.

4.2 The successful bidder shall furnish for Government approval, within 30 days after award of the contract, the name of the manufacturer of the fire alarm system he proposes to furnish, together with an official written statement from the said manufacturer adequately describing the system, and giving name and catalog number or other identification with a brief description of each part thereof. The manufacturer shall also state what commercial approval, if any, has been accorded his proposed system.

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4.2.1 After approval of the name of the manufacturer of the system, the successful bidder shall furnish for Government approval complete outline drawings in quadruplicate, unless otherwise specified, of the stations, alarm bells, flush back boxes, recording equipment, control unit, front and rear elevation of the control board; a full schematic wiring diagram, and a wiring diagram of all circuits external to the panel from which a conduit layout can be made; and, unless otherwise specified, four copies of an instruction manual completely describing the operation and maintenance of the system.

4.2.2 On request, the successful bidder shall submit for approval and test by a Government laboratory, a complete working system consisting of the necessary stations and signalling devices, and a control unit containing at least one station and two alarm bell circuits.

4.3 **Sampling.** If any department or agency requires tests in addition to the statements and certified reports referred to in 3.2 and 4.1.1, one sample of each system component involved shall be supplied to the testing laboratory.

4.4 **Inspection.** Each component shall be subjected to a thorough examination to ascertain that the material, workmanship, and design are in accordance with the requirements of this specification.

4.5 **Tests.** Unless otherwise specified (see 6.1), components selected in accordance with 4.3, shall be subjected to the tests necessary to determine compliance with the requirements of 3.14.1, 3.14.2 and/or 3.14.3. The tests shall be conducted as specified in the applicable UL Standard or section of NEMA Standard SB1-1962.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the various levels of packaging and packing shall be as specified in Fed. Std. No. 102.)

5.1 **Packaging.**

5.1.1 *Level C.* Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 **Packing.**

5.2.1 *Level C.* The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.
(*b*) Class, style, and kind of system (1.2.1.1).
(*c*) Class and style of station (1.2.2.1).
(*d*) Type of automatic detectors (1.2.3.1).
(*e*) Single-stroke or vibrating bells (3.3.1).
(*f*) Other signalling devices or systems, if required (3.3.1).

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(*g*) Power supply and voltage required (3.3.2.1).
(*h*) Evacuation alarm, if required (3.3.3.1.1 and 3.3.4.1.1).
(*i*) General alarm switch and additional signals, if required (3.3.4.2).
(*j*) Surface mounted stations, if required (3.4.3.2).
(*k*) Station finish, if other than specified (3.4.3.2.2).
(*l*) Special purpose station material and finish (3.4.3.4).
(*m*) Alarm bell finish, if other than specified (3.7.2.2).
(*n*) Alarm bell diameters, if other than specified (3.7.2.3).
(*o*) Flush mounting of alarm bells, if required (3.7.2.4).
(*p*) Type of audible trouble signal, if other than specified (3.7.3).
(*q*) Type of control board mounting (3.8.3).
(*r*) Rear door cabinets, if required (3.8.4.2).
(*s*) Number of circuits, loops, stations, detectors, transmitters, and alarm bells required (3.9).
(*t*) Automatic or manual operation on open or grounded circuit, if required (3.9.3).
(*u*) Graphic annunciation, if required (3.11).
(*v*) Performance requirements, if other than specified (3.14).
(*w*) Inspection responsibility, if other than specified (4.1).
(*x*) Number of outline drawings and instruction manuals required, if other than specified (4.2.1).
(*y*) Approval test of sample system, if required (4.2.2).
(*z*) Additional component testing, if required (4.3).
(*aa*) Component tests, if other than specified (4.5).
(*bb*) Special marking, if required (5.3).

6.2 It is within the province of the drawings and project specification to define the methods of installation, finish of apparatus, and minor modifications of details to adapt this specification to special conditions which may prevail for any particular project, but no authority is granted to waive the general requirements, or to modify the principles of the system or apparatus.

6.3 For the convenience of bidders, it is recommended that the number and locations of the stations and alarm bells be summarized in the invitation for bids.

6.4 The attention of building designers and engineers is directed to the fact that, when selecting automatic detectors for use in style B systems, care must be exercised since, in locations where ambient temperature conditions are subject to rapid changes (such as caused by heavy-duty blower heaters, operation of oven doors, etc.) rate-of-rise detectors may be unsuitable. In such applications, the fixed temperature type should be used to reduce the possibility of false alarms.

6.5 To provide for proper system operation, it is suggested that a maximum of 25 stations, transmitters, or a combination of the two be provided in any one alarm initiating loop and that no more than 10 alarm bells be installed in series in any one alarm indicating loop.

6.6 In some installations, it may be desirable to include a number of prewound coded stations. When such is the case, these stations shall provide a noncoded audible trouble signal when in the rundown condition.

6.7 Contract provisions for fire alarm systems furnished under this specification should require that wiring in conduit be limited to 60 percent fill and that junction boxes with terminal blocks be installed at the base of all vertical risers to facilitate installation and maintenance of the system.

6.8 **Transportation description.** The transportation description and minimum weights applicable to this commodity are:

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Rail:

Fire alarm boxes, electric
Carload minimum weight—30,000 pounds.

Motors:

Fire alarm boxes, electric
Truckload minimum weight—30,000 pounds subject to Rule 115 National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:

Army—MO MI

Navy—DOCKS

Copies of this specification may be purchased for 10 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

PECIFICATION

RGANIZATION *(Of submitter)* | CITY AND STATE

ONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

ATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

IBMITTED BY *(Printed or typed name and activity)* | DATE

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